Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/17/2008
PI Name:	Raikhel, Natasha Ph.D.		
Project Title:	An understanding of gravitropism via vacuole biogensis and the identification of gravi-active drugs to dissect the pathway		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	02/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hicks, Glenn (University of California)		
Grant/Contract No.:	NNA04CI26G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Understanding the effects of and devising countermeasures to microgravity, radiation and other environmental factors (vibration, altered atmosphere) are critical to long-term spaceflight. An in-depth understanding of plant perception and responses to these challenges, particularly microgravity, will be essential for the development of life support systems, the production of food and for the psychological well-being of astronauts. There is a substantial body of literature concerning the perception and responses of plants to gravity yet the process is poorly understood at the molecular and cellular level. Recent evidence has established a link between endomembrane trafficking, vacuole biogenesis, auxin transport and gravitropism. We developed a screen for mutants defective in vacuole biogenesis in Arabidopsis lines expressing a tonoplast marker (GFP::?TIP). The marker permits the vacuole to be visualized by confocal microscopy. Representatives from at least two mutant classes display defects in shoot gravitropism. Our NASA-funded research leverages these and other endomembrane mutants to understand mechanistic details of the gravitropic pathway. Another aspect of our research is chemical library screening to identify compounds that affect gravitropic perception, signaling or response. This approach presents an exciting new avenue towards expanding our insight into mechanisms underlying gravitropism and in defining potential pharmacological countermeasures to negative effects of microgravity. During year one of NASA funding we have characterized and mapped to low resolution two novel gravitropic mutants that are defective in vacuole biogenesis. Fine mapping and gene cloning are proceeding. We have also completed an ambitious screen of a 10,000 compound chemical library and have identified at least three unique compounds (non-auxins) that inhibit gravitropism in Arabidopsis seedlings with additional bioactive compounds anticipated. Identification of the gene targets will begin in the coming via genetic screens for mutants that are resistant or hypersensitive to our gravi-active compounds. These results should provide new insights into mechanisms of gravity sensing, signal transduction and response in support of NASA ground-based and long-term spaceflight objectives. Our goal is to understand mechanisms of gravity perception, signaling and response by leveraging novel mutants that are defective in both gravitropism and endomembrane pathways and by identifying novel chemicals (and their cognate gene targets) that inhibit or enhance gravitropism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Earth benefits are derived in several ways: 1. The ability of plants (and animals) to orient themselves in response to gravity is essential for proper development. Development is in turn a complex program involving multiple cellular pathways including hormones, gene regulation and vesicular transport. Gravitropism intersects these pathways. Thus its study will provide insights into complex developmental pathways that may result in improved growth habit and nutritional value of food crops. 2. Our mutants and chemicals affect both gravitropism and endomembrane and vacuole biogenesis. All major seed crops such as maize, rice, wheat, soy and canola depend upon the deposition of proteins, lipids or starch in specialized vacuoles. Our research will provide insight into vacuole biogenesis which may lead to the improved nutritional value and shelf life of seeds. This has potential multibillion dollar vaue to the US economy.
Task Progress:	Overall, our NASA funding allowed us to characterize EMS mutants that are defective in gravitropism. We successfully fine-mapped agg1and discovered that it is an unexpected protein involved in gravitropism and vacuole biogenesis. These are exciting results and NASA funding has permitted us to make significant progress. In addition we have been able to use chemical genomics to identify a series of compounds that affect endomembrane trafficking and gravitropism. During the period of funding we successfully identified one of the targets of Gravicin and found it to be a member of the PGP family of auxin transporters. Thus, we have a novel antagonist of auxin transport that also appears to have a second target which impacts endomembrane trafficking. We are continuing to pursue this and our other compounds for novel activities and targets at the intersection of gravitropism and trafficking with a long-term goal of understanding the link between these fundamental biological pathways. Even with a cut in the year three award, this work was facilitated greatly by NASA funding. We hope that in the near future NASA will have the resources to again invest in such fundamental biological sciences that impact its long-term goals of sustained habitation in space.
Bibliography Type:	Description: (Last Updated: 06/02/2014)
Articles in Peer-reviewed Journals	Rojas-Pierce M, Titapiwatanakun B, Sohn EJ, Fang F, Larive CK, Blakeslee J, Cheng Y, Cuttler S, Peer WA, Murphy AS, Raikhel NV. "Arabidopsis P-glycoprotein19 participates in the inhibition of gravitropism by gravacin." Chem Biol. 2007 Dec;14(12):1366-76. PMID: 18096505 , Dec-2007
Articles in Peer-reviewed Journals	Surpin M, Rojas-Pierce M, Carter C, Hicks GR, Vasquez J, Raikhel NV. "The power of chemical genomics to study the link between endomembrane system components and the gravitropic response." Proc Natl Acad Sci U S A. 2005 Mar 29;102(13):4902-7. (Erratum in: Proc Natl Acad Sci U S A. 2005 Jul 26;102(30):10752.) PMID: 15772170 , Jul-2005
Articles in Peer-reviewed Journals	Girke T, Cheng LC, Raikhel N. "ChemMine. A compound mining database for chemical genomics." Plant Physiol. 2005 Jun;138(2):573-7. PMID: 15955920 , Jun-2005
Articles in Peer-reviewed Journals	Surpin M. "Plant chemical genomics: gravity sensing and response." Methods Mol Biol. 2014;1056:115-24. http://dx.doi.org/10.1007/978-1-62703-592-7_12 ; PubMed PMID: 24306868 , Jan-2014